Weichen (Arthur) Zhou, PhD

Research Assistant Professor
LEO lecturer
University of Michigan Medical School, Gilbert S. Omenn Department of Computational Medicine & Bioinformatics
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Education and Training

Education

09/2005-07/2009	B.S.E., Bioinformatics, Huazhong University of Science and Technology, Wuhan, China
09/2009-07/2014	Ph.D., Genetics, Fudan University, Shanghai, China

Postdoctoral Training

11/2015-09/2020	Research Fellow, Department of Computational Medicine & Bioinformatics, University of Michigan, Ann Arbor, MI

Work Experience

Academic Appointment

10/2020-08/2024	Research Investigator, Department of Computational Medicine and Bioinformatics, University of Michigan, Ann Arbor, MI
08/2021-Present	LEO Lecturer in Bioinformatics, Department of Computational Medicine and Bioinformatics, University of Michigan, Ann Arbor, MI
09/2024-Present	Research Assistant Professor, Department of Computational Medicine & Bioinformatics, University of Michigan, Ann Arbor, Michigan

Research Position

07/2014-10/2015	Research Associate, Ministry of Education Key Laboratory of Contemporary Anthropology, Fudan University, Shanghai

Grants

Current Grants

R01DE034786:*Revealing the Trajectory and Critical Roles of Retinoic Acid in Ameloblast Differentiation:*
MPI
Tian Liang(MPI)
NIH-DHHS-US
03/2026 - 02/2028
$312,000

R03DE034786:*Revealing the Trajectory and Critical Roles of Retinoic Acid in Ameloblast Differentiation:*
Funded by
National Institute of Dental and Craniofacial Research
03/2026 - 02/2028
$312,000

R03AG087485:*Leveraging large datasets to determine genomic impact of transposable elements in Alzheimers Disease and related dementias:*
PI
NIH-DHHS-US
06/2025 - 05/2027
$303,449

Submitted - Open

R35:*The Impact of Transposable Element Heterogeneity in Humans:*
PI
NIH-DHHS-US
07/2026 - 06/2031
$2,145,000

R01NS145291:*Characterization and functional impact of somatic numtogenesis in the human cortex:*
Co-I (Principal Investigator:Ryan Mills)
NIH-DHHS-US
04/2026 - 03/2031
$3,809,730

Past Grants

P30AG072931:*Leverage large datasets to determine the impact of nuclear mitochondrial DNA insertion on Alzheimer's disease and related dementias:*
PI
NIH/NIA-funded Michigan Alzheimer's Disease Research Center
07/2023 - 06/2025
$25,000

P30AG072931:*Explore the functional impact of transposable elements in Alzheimer's disease and related dementias:*
PI
NIH/NIA-funded Michigan Alzheimer's Disease Research Center
07/2022 - 07/2024
$50,000

3R21HG011493-02S1:*New technologies for accurate capture and sequencing of repeat-associated regions:*
Co-I
National Institute on Aging
06/2022 - 11/2022
$230,098

N/A:*The 4th Research Funding of National Key Disciplines:*
Funded by
Ministry of Education of People's Republic of China
09/2013 - 07/2014
CNY 42,000

Honors and Awards

International

2026 - Present	Alzheimer's Association International Conference Fellowship Award, Alzheimer's Association International Conference 2026

National

2019	Charles J. Epstein Trainee Awards for Excellence in Human Genetics Research, semifinalist, the 69th Annual Meeting of the American Society of Human Genetics, the American Society of Human Genetics

Institutional

2010	Academic Scholarships, Fudan University, Fudan University, China
2011	Academic Scholarships, Fudan University, Fudan University, China
2012	Academic Scholarships, Fudan University, Fudan University, China
2013	National Scholarship for Graduate Students, Ministry of Education of PRC, China
2013	Outstanding Graduate Students, Fudan University, Fudan University, China
2014	Shanghai Outstanding Graduates, Shanghai Municipal Education Commission, China
2016	Outstanding Dissertation of Ph.D. (2015), Shanghai Municipal Education Commission, China
2019	Office of Graduate Postdoctoral Studies (OGPS) Postdoctoral Travel Award, University of Michigan Medical School
2024	Basic Sciences 2024 Research Staff Award, University of Michigan Medical School, United States
2026 - Present	Michigan Alzheimer's Disease Center's Isadore & Margaret Mezey Junior Investigator Conference Award, Michigan Alzheimer's Disease Center

Study Sections, Editorial Boards, Journal & Abstract Review

Editorial Boards / Journal & Abstract Reviews

Editorial Boards

2020 - Present	Editorial Reviewer Board, Frontiers in Genetics
2020 - Present	Editorial Reviewer Board, Frontiers in Bioengineering and Biotechnology

Journal Review

2018	Genome Research (Ad Hoc)
2019	Nucleic Acids Research (Ad Hoc)
2020	American Journal of Human Genetics (Ad Hoc)
2020	Nature Communications (Ad Hoc)
2020	Molecular Biology Report (Ad Hoc)
2020	Computational Biology and Chemistry (Ad Hoc)
2020	PLoS ONE (Ad Hoc)
2020	Frontiers in Genetics (Ad Hoc)
2021	Cell Genomics (Ad Hoc)
2021	GigaScience (Ad Hoc)
2021 - 2022	Bioinformatics (Ad Hoc)
2023	Genome Biology (Ad Hoc)
2023	Genome Biology (Ad Hoc)
2024	Nature Reviews Genetics (Ad Hoc)

2024	Bioinformatics (Ad Hoc)
2024	Biochemical Society Transactions (Ad Hoc)
2024	Computers in Biology and Medicine (Ad Hoc)
2024	Bioinformatics (Ad Hoc)
2024	Nature Methods (Ad Hoc)
2025	Nature Methods (Ad Hoc)
2025	Nature Communications (Ad Hoc)
2025	Genome Biology (Ad Hoc)
2025	Nature Communications (Ad Hoc)
2026	Nature Communications (Ad Hoc)

Teaching

Mentorship

Graduate Students – Master's

09/2022 - 09/2025	Yanming Gan, Harvard Medical School, Research mentorship. Ph.D. Program at the University of Chicago
09/2025 - 01/2026	Wenjie Han, University of Michigan, School of Public Health, Research mentorship

Postbaccalaureate

05/2023 - 08/2023	Sophia Marcotte, Michigan Institute for Clinical & Health Research, MICHR Summer Research Program, Mentoring graduate students gaining hands-on research experience in wide range of topics. Career mentorship

Undergraduate Student

09/2021 - 08/2022	Yanming Gan, University of Michigan, Research mentorship. Master Program in Biomedical Informatics, Harvard Medical School
07/2022 - 08/2023	Sophia Marcotte, University of Michigan, Research mentorship. Mentee in MICHR Summer Research Program 2023. Master Program in Biomedical Informatics, Boston University

Visiting Scholar

07/2019 - 09/2019	Thomas Chang, University of Michigan, Research mentorship
07/2019 - 09/2019	Steven Kim, University of Michigan, Research mentorship
02/2023 - 09/2024	Irfan Darwish, University of Michigan Medical School, Department of Computational Medicine and Bioinformatics, Research mentorship

Teaching Activity

Institutional

09/2010 - 01/2011	Journal club of Human Evolution, Fudan University, Teaching Assistant
02/2011 - 06/2011	Human Evolutionary Genetics, Fudan University, Teaching Assistant
02/2012 - 06/2012	Human Evolutionary Genetics, Fudan University, Teaching Assistant
09/2012 - 01/2013	Human Evolution, Fudan University, Teaching Assistant
02/2013 - 06/2013	Human Evolutionary Genetics, Fudan University, Teaching Assistant
08/2021 - 12/2021	BIOINF 527 Introduction to Bioinformatics and Computational Biology, University of Michigan Medical School, Department of Computational Medicine and Bioinformatics, Lecturer

08/2022 - 12/2022	BIOINF 527 Introduction to Bioinformatics and Computational Biology, University of Michigan Medical School, Department of Computational Medicine & Bioinformatics, Lecturer
08/2023 - 12/2023	BIOINF 527 Introduction to Bioinformatics and Computational Biology, University of Michigan Medical School, Department of Computational Medicine & Bioinformatics, Lecturer
08/2024 - 12/2024	BIOINF 527 Introduction to Bioinformatics and Computational Biology, University of Michigan Medical School, Computational Medicine and Bioinformatics, Lecturer
09/2025 - 12/2025	BIOINF 527 Introduction to Bioinformatics and Computational Biology, University of Michigan Medical School, Department of Computational Medicine & Bioinformatics, Lecturer
02/2026 - 05/2026	BIOINF 529 Bioinformatics Concepts and Algorithms, University of Michigan Medical School, Department of Computational Medicine & Bioinformatics, Lecturer

Memberships in Professional Societies

2016 - Present	Member, The American Society of Human Genetics
2017 - Present	Member, The Association of Chinese Geneticists in America
2023 - Present	Member, The Alzheimer's Association International Society to Advance Alzheimer's Research and Treatment
2023 - Present	Member, The National Center for Faculty Development & Diversity

Volunteer Service

Volunteer

2018	Judge, Forsythe Science Expo, Ann Arbor, Forsythe Science Expo, Ann Arbor
2020	Volunteer, The American Society of Human Genetics 2020, ASHG Connect and Shared Interest Groups, the American Society of Human Genetics
2021	Career mentorship, The American Society of Human Genetics 2021, Mentor of Career Paths in Genetics Networking Session: Perfect Your Elevator Speech, Professional Development and Industry Technology Forum, The American Society of Human Genetics
2025	Abstract Reviewer, The American Society of Human Genetics 2025, Abstract Reviewer
2025	The American Society of Human Genetics 2025, Platform Session Moderator

Scholarly Activities

Presentations

Extramural Invited Presentation

Speaker

1. Whole genome functional traits and evolutionary clues between Copy Number Variations and Segmental Duplications in human genome, **Zhou W**, The Annual Meeting of Shanghai Genetics Society, 07/2013, Shanghai

2. Refining polymorphic retrotransposon insertions in human genomes, **Zhou W**, American Society of Human Genetics, 10/2019, Texas, Huston

3. Comprehensive mobile element variation and effects across 64 genetically diverse haplotypes, **Zhou W**, The Cold Spring Harbor Meeting: Transposable Elements, 05/2022, Cold Spring Harbor

4. Establishing a benchmark for bulk and single-cell somatic genomic analysis using diverse technologies, **Zhou W**, Somatic Mosaicism across Human Tissues, 12/2023, US

5.   Explore the potential impact of cryptic genomic insertions in Alzheimer's Disease, **Zhou W**, Michigan Alzheimer's Disease Center, 06/2024, MSU College of Human Medicine - Secchia Center, Grand Rapid, Michigan

6.   A personalized multi-platform assessment of somatic mosaicism in the human frontal cortex, **Zhou W**, American Society of Human Genetics, 11/2024, Denver, Colorado

Moderator

1.   Function of Complex Variation and Repeats, American Society of Human Genetics, 10/2025, Boston, MA

Intramural Invited Presentation

Speaker

1.   PALMER: Identification and characterization of occult mobile element insertions using long-read sequencing technology, **Zhou W**, University of Michigan, DCMB Tools and Technology Seminar Series, 03/2020, Ann Arbor

2.   Somatic nuclear mitochondrial DNA insertions are prevalent in the human brain and accumulate over time in fibroblasts, **Zhou W**, Michigan Alzheimer's Disease Center, Leaders Initiative Seminar Series, 01/2025, Virtual

3.   A personalized multi-platform framework enhances somatic mosaicism discovery in human tissues, **Zhou W**, Michigan George O'Brien Kidney Center, 11/2025, Ann Arbor, Michigan

Publications/Scholarship

(Co-First Author *; Corresponding author **; Co-Last author ***)

Peer Reviewed

Journal Article

1.   **Zhou W**, Tan J: Dental Anthropology Suggests Southeast Asian Origins amongst the Jomon People of Japan. *Communication on Contemporary Anthropology*.04(01)06/2010. DOI:10.4236/coca.2010.41012

2.   **Zhou W**: Genetic and environmental factors of pathogeny: a population genetics view. *Communication on Contemporary Anthropology*.04(01)12/2010. DOI:10.4236/coca.2010.41021

3.   Lin R, Fu W, **Zhou W**, Wang Y, Wang X, Huang W, Jin L: Association of heme oxygenase-1 gene polymorphisms with essential hypertension and blood pressure in the Chinese Han population. *Genetic Testing and Molecular Biomarkers*.15(1-2): 23-28, 01/2011. PM21275653

4.   Lin R, Wang X, **Zhou W**, Fu W, Wang Y, Huang W, Jin L: Association of Polymorphisms in the Solute Carrier Organic Anion Transporter Family Member 1B1 Gene with Essential Hypertension in the Uyghur Population. *Annals of Human Genetics*.75(2): 305-311, 03/2011. PM21309757

5.   Lin R, Wang X, **Zhou W**, Fu W, Wang Y, Huang W, Jin L: Association of a BLVRA common polymorphism with essential hypertension and blood pressure in Kazaks. *Clinical and Experimental Hypertension*.33(5): 294-298, 08/2011. PM21721974

6.   **Zhou W**, Zhang F, Chen X, Shen Y, Lupski JR, Jin L: Increased genome instability in human DNA segments with self-chains: Homology-induced structural variations via replicative mechanisms. *Human Molecular Genetics*.22(13): 2642-2651, 07/2013. PM23474816

7.   Chen L, **Zhou W**, Zhang L, Zhang F: Genome Architecture and Its Roles in Human Copy Number Variation. *Genomics & Informatics*.12(4): 136-136, 01/2014. PM25705150

8.   Chen Y, Guo L, Chen J, Zhao X, **Zhou W**, Zhang C, Wang J, Jin L, Pei D, Zhang F: Genome-wide CNV analysis in mouse induced pluripotent stem cells reveals dosage effect of pluripotent factors on genome integrity. *BMC Genomics*.15(1): 79, 01/2014. PM24472662

9. Wu N, Ming X, Xiao J, Wu Z, Chen X, Shinawi M, Shen Y, Yu G, Liu J, Xie H, Gucev ZS, Liu S, Yang N, Al-Kateb H, Chen J, Zhang J, Hauser N, Zhang T, Tasic V, Liu P, Su X, Pan X, Liu C, Wang L, Shen J, Shen J, Chen Y, Zhang T, Zhang J, Choy KW, Wang J, Wang Q, Li S, **Zhou W**, Guo J, Wang Y, Zhang C, Zhao H, An Y, Zhao Y, Wang J, Liu Z, Zuo Y, Tian Y, Weng X, Sutton VR, Wang H, Ming Y, Kulkarni S, Zhong TP, Giampietro PF, Dunwoodie SL, Cheung SW, Zhang X, Jin L, Lupski JR, Qiu G, Zhang F: TBX6 null variants and a common hypomorphic allele in congenital scoliosis. *New England Journal of Medicine*.372(4): 341-350, 01/2015. PM25564734

10. Peng Z, **Zhou W**, Fu W, Du R, Jin L, Zhang F: Correlation between frequency of non-allelic homologous recombination and homology properties: Evidence from homology-mediated CNV mutations in the human genome. *Human Molecular Genetics*.24(5): 1225-1233, 03/2015. PM25324539

11. Chen L, **Zhou W**, Zhang C, Lupski JR, Jin L, Zhang F: CNV instability associated with DNA replication dynamics: Evidence for replicative mechanisms in CNV mutagenesis. *Human Molecular Genetics*.24(6): 1574-1583, 03/2015. PM25398944

12. McConnell MJ, Moran JV, Abyzov A, Akbarian S, Bae T, Cortes-Ciriano I, Erwin JA, Fasching L, Flasch DA, Freed D, Ganz J, Jaffe AE, Kwan KY, Kwon M, Lodato MA, Mills RE, Paquola AC M, Rodin RE, Rosenbluh C, Sestan N, Sherman MA, Shin JH, Song S, Straub RE, Thorpe J, Weinberger DR, Urban AE, Zhou B, Gage FH, Lehner T, Senthil G, Walsh CA, Chess A, Courchesne E, Gleeson JG, Kidd JM, Park PJ, Pevsner J, Vaccarino FM: Intersection of diverse neuronal genomes and neuropsychiatric disease: The Brain Somatic Mosaicism Network. *Science*.356(6336)04/2017. PM28450582

13. Zhang L, Wang J, Zhang C, Li D, Carvalho CM B, Ji H, Xiao J, Wu Y, **Zhou W**, Wang H, Jin L, Luo Y, Wu X, Lupski JR, Zhang F, Jiang Y: Efficient CNV breakpoint analysis reveals unexpected structural complexity and correlation of dosage-sensitive genes with clinical severity in genomic disorders. *Human Molecular Genetics*.26(10): 1927-1941, 05/2017. PM28334874

14. **Zhou W**, Ma Y, Zhang J, Hu J, Zhang M, Wang Y, Li Y, Wu L, Pan Y, Zhang Y, Zhang X, Zhang X, Zhang Z, Zhang J, Li H, Lu L, Jin L, Wang J, Yuan Z, Liu J: Predictive model for inflammation grades of chronic hepatitis B: Large-scale analysis of clinical parameters and gene expressions. *Liver Int*.37(11): 1632 -1641, 11/2017. PM28328162

15. **Zhou W**, Wang Y, Fujino M, Shi L, Jin L, Li XK, Wang J: A standardized fold change method for microarray differential expression analysis used to reveal genes involved in acute rejection in murine allograft models. *FEBS Open Bio*.8(3): 481-490, 03/2018. PM29511625

16. Li Y, Liu X, Ma Y, Wang Y, **Zhou W**, Hao M, Yuan Z, Liu J, Xiong M, Shugart YY, Wang J, Jin L: knnAUC: an open-source R package for detecting nonlinear dependence between one continuous variable and one binary variable. *BMC Bioinformatics*.19(1): 448, 11/2018. PM30466390

17. **Zhou W**, Emery SB, Flasch DA, Wang Y, Kwan KY, Kidd JM, Moran JV, Mills RE: Identification and characterization of occult human-specific LINE-1 insertions using long-read sequencing technology. *Nucleic Acids Research*.48(3): 1146-1163, 01/2020. PM31853540

18. Zook JM, Hansen NF, Olson ND, Chapman L, Mullikin JC, Xiao C, Sherry S, Koren S, Phillippy AM, Boutros PC, Sahraeian SM E, Huang V, Rouette A, Alexander N, Mason CE, Hajirasouliha I, Ricketts C, Lee J, Tearle R, Fiddes IT, Barrio AM, Wala J, Carroll A, Ghaffari N, Rodriguez OL, Bashir A, Jackman S, Farrell JJ, Wenger AM, Alkan C, Soylev A, Schatz MC, Garg S, Church G, Marschall T, Chen K, Fan X, English AC, Rosenfeld JA, **Zhou W**, Mills RE, Sage JM, Davis JR, Kaiser MD, Oliver JS, Catalano AP, Chaisson MJ P, Spies N, Sedlazeck FJ, Salit M: A robust benchmark for detection of germline large deletions and insertions. *Nature Biotechnology*.38(11): 1347-1355, 11/2020. PM32541955

19. Dayama G, **Zhou W**, Prado-Martinez J, Marques-Bonet T, Mills RE: Characterization of nuclear mitochondrial insertions in the whole genomes of primates. *Nar Genomics and Bioinformatics*.2(4): lqaa089, 12/2020. PM33575633

20. Oleksyk TK, Wolfsberger WW, Weber AM, Shchubelka K, Oleksyk OT, Levchuk O, Patrus A, Lazar N, Castro-Marquez SO, Hasynets Y, Boldyzhar P, Neymet M, Urbanovych A, Stakhovska V, Malyar K, Chervyakova S, Podoroha O, Kovalchuk N, Rodriguez-Flores JL, **Zhou W**, Medley S, Battistuzzi F, Liu R, Hou Y, Chen S, Yang H, Yeager M, Dean M, Mills RE, Smolanka V: Genome diversity in Ukraine. *Gigascience*.10(1)01/2021. PM33438729

21. Zhu X, Zhou B, Pattni R, Gleason K, Tan C, Kalinowski A, Sloan S, Fiston-Lavier AS, Mariani J, Petrov D, Barres BA, Duncan L, Abyzov A, Vogel H, Urban A, Walsh C, Ganz J, Woodworth M, Li P, Rodin R, Hill R, Bizzotto S, Zhou Z, Lee A, D'Gama A, Galor A, Bohrson C, Kwon D, Gulhan D, Lim E, Cortes I, Luquette J, Sherman M, Coulter M, Lodato M, Park P, Monroy R, Kim S, Dou Y, Chess A, Jones A, Rosenbluh C, Akbarian S, Langmead B, Thorpe J, Pevsner J, Scharpf R, Cho S, Vaccarino F, Fasching L, Tomasi S, Sestan N, Pochareddy S, Jaffe A, Paquola A, Weinberger D, Erwin J, Shin J, Straub R, Narurkar R, Addington A, Panchision D, Meinecke D, Senthil G, Bingaman L, Dutka T, Lehner T, Abyzov A, Bae T, Saucedo-Cuevas L, Conniff T, Flasch DA, Frisbie TJ, Kidd JM, Lam MM, Moldovan JB, Moran JV, Kwan KY, Mills RE, Emery S, **Zhou W**, Wang Y, Daily K, Peters M, Gage F, Wang M, Reed P, Linker S, Sarkar A, Serres A, Juan D, Povolotskaya I, Lobon I, Solis M, Garcia R, Marques-Bonet T, Mathern G, Courchesne E, Gu J: Machine learning reveals bilateral distribution of somatic L1 insertions in human neurons and glia. *Nature Neuroscience*.24(2): 186-196, 02/2021. PM33432196

22. Rodin RE, Dou Y, Kwon M, Sherman MA, D'Gama AM, Doan RN, Rento LM, Girskis KM, Bohrson CL, Kim SN, Nadig A, Luquette LJ, Gulhan DC, Walsh CA, Ganz J, Woodworth MB, Li P, Hill RS, Bizzotto S, Zhou Z, Lee EA, Barton AR, Galor A, Kwon D, Lim ET, Cortes IC, Coulter ME, Lodato MA, Park PJ, Monroy RB, Dou Y, Chess A, Gulyás-Kovács A, Rosenbluh C, Akbarian S, Ben Langmead, Thorpe J, Pevsner J, Cho S, Jaffe AE, Paquola A, Weinberger DR, Erwin JA, Shin JH, Straub RE, Narurkar R, Abyzov AS, Bae T, Addington A, Panchision D, Meinecke D, Senthil G, Bingaman L, Dutka T, Lehner T, Saucedo-Cuevas L, Conniff T, Daily K, Peters M, Gage FH, Wang M, Reed PJ, Linker SB, Urban AE, Zhou B, Zhu X, Serres A, Juan D, Povolotskaya I, Lobón I, Solis-Moruno M, García-Pérez R, Marquès-Bonet T, Mathern GW, Courchesne E, Gu J, Gleeson JG, Ball LL, George RD, Pramparo T, Flasch DA, Frisbie TJ, Kidd JM, Moldovan JB, Moran JV, Kwan KY, Mills RE, Emery SB, **Zhou W**, Wang Y, Ratan A, McConnell MJ, Vaccarino FM: The landscape of somatic mutation in cerebral cortex of autistic and neurotypical individuals revealed by ultra-deep whole-genome sequencing. *Nature Neuroscience*.24(2): 176-185, 02/2021. PM33432195

23. Ebert P, Audano PA, Zhu Q, Rodriguez-Martin B, Porubsky D, Bonder MJ, Sulovari A, Ebler J, **Zhou W**, Mari RS, Yilmaz F, Zhao X, Hsieh PH, Lee J, Kumar S, Lin J, Rausch T, Chen Y, Ren J, Santamarina M, Höps W, Ashraf H, Chuang NT, Yang X, Munson KM, Lewis AP, Fairley S, Tallon LJ, Clarke WE, Basile AO, Byrska-Bishop M, Corvelo A, Evani US, Lu TY, Chaisson MJ P, Chen J, Li C, Brand H, Wenger AM, Ghareghani M, Harvey WT, Raeder B, Hasenfeld P, Regier AA, Abel HJ, Hall IM, Flicek P, Stegle O, Gerstein MB, Tubio JM C, Mu Z, Li YI, Shi X, Hastie AR, Ye K, Chong Z, Sanders AD, Zody MC, Talkowski ME, Mills RE, Devine SE, Lee C, Korbel JO, Marschall T, Eichler EE: Haplotype-resolved diverse human genomes and integrated analysis of structural variation. *Science*.372(6537)04/2021. PM33632895

24. Zhao X, Collins RL, Lee W-P, Weber AM, Jun Y, Zhu Q, Weisburd B, Huang Y, Audano PA, Wang H, Walker M, Lowther C, Fu J, Human Genome Structural Variation Consortium, Gerstein MB, Devine SE, Marschall T, Korbel JO, Eichler EE, Chaisson MJ P, Lee C, Mills RE, Brand H, Talkowski ME: Expectations and blind spots for structural variation detection from long-read assemblies and short-read genome sequencing technologies. *Am J Hum Genet*.108(5): 919-928, 05/2021. PM33789087

25. Wang Y, Bae T, Thorpe J, Sherman MA, Jones AG, Cho S, Daily K, Dou Y, Ganz J, Galor A, Lobon I, Pattni R, Rosenbluh C, Tomasi S, Tomasini L, Yang X, Zhou B, Akbarian S, Ball LL, Bizzotto S, Emery SB, Doan R, Fasching L, Jang Y, Juan D, Lizano E, Luquette LJ, Moldovan JB, Narurkar R, Oetjens MT, Rodin RE, Sekar S, Shin JH, Soriano E, Straub RE, **Zhou W**, Chess A, Gleeson JG, Marquès-Bonet T, Park PJ, Peters MA, Pevsner J, Walsh CA, Weinberger DR, Vaccarino FM, Moran JV, Urban AE, Kidd JM, Mills RE, Abyzov A: Comprehensive identification of somatic nucleotide variants in human brain tissue. *Genome Biology*.22(1)12/2021. PM33781308

26. McDonald TL, **Zhou W**, Castro CP, Mumm C, Switzenberg JA, Mills RE, Boyle AP: Cas9 targeted enrichment of mobile elements using nanopore sequencing. *Nature Communications*.12(1): 3586, 12/2021. PM34117247

27. Bao Y, Wadden J, Erb-Downward JR, Ranjan P, **Zhou W**, McDonald TL, Mills RE, Boyle AP, Dickson RP, Blaauw D, Welch JD: SquiggleNet: real-time, direct classification of nanopore signals. *Genome Biology*.22(1): 298, 12/2021. PM34706748

28. Lin J, Yang X, Kosters W, Xu T, Jia Y, Wang S, Zhu Q, Ryan M, Guo L, Gerstein MB, Sanders AD, Zody MC, Talkowski ME, Mills RE, Korbel JO, Marschall T, Ebert P, Audano PA, Rodriguez-Martin B, Porubsky D, Jan Bonder M, Sulovari A, Ebler J, **Zhou W**, Serra Mari R, Yilmaz F, Zhao X, Hsieh P, Lee J, Kumar S, Rausch T, Chen Y, Chong Z, Munson KM, Chaisson MJ P, Chen J, Shi X, Wenger AM, Harvey WT, Hansenfeld P, Regier A, Hall IM, Flicek P, Hastie AR, Fairely S, Zhang C, Lee C, Devine SE, Eichler EE, Ye K: Mako: A Graph-Based Pattern Growth Approach to Detect Complex Structural Variants. *Genomics, Proteomics & Bioinformatics*.20(1): 205-218, 02/2022. PM34224879

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